

Relaxation and Odd-Even Effect in Fermi Liquids

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- k -space
- Fermi Dirac distribution
- Boltzmann equation

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Recovering equations

- Linearization of the collision integral

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Introduction

1.1

***k*-space**

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k -space to real space

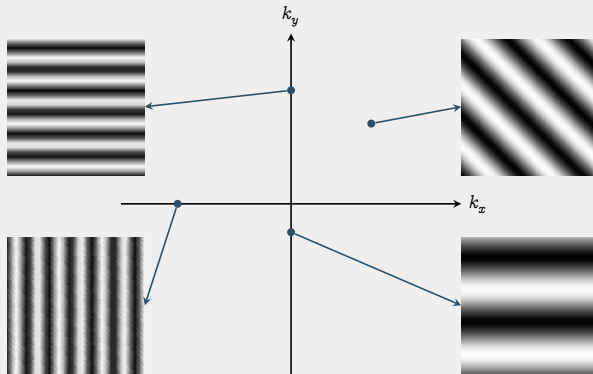


Figure 1: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

1.2

Fermi Dirac distribution

Fermi-Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

1

- $\beta = 1/k_B T$ is the inverse temperature
- $\varepsilon(\mathbf{p}) = \frac{\hbar^2 \mathbf{p}^2}{2m}$ is the energy of particles with momentum $\mathbf{p}$
- μ is the chemical potential

The Fermi-Dirac distribution is in **equilibrium**.

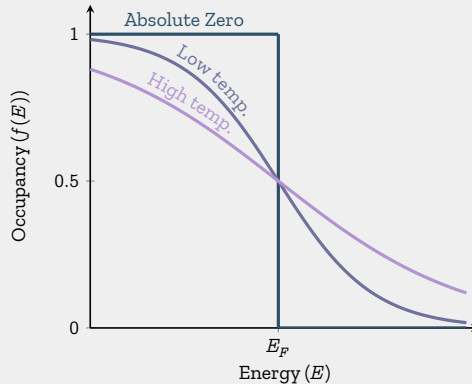


Figure 2: FERMI-DIRAC DISTRIBUTION AT DIFFERENT TEMPERATURES

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

The Fermi surface is the set of points in k -space where the energy of the electrons is equal to the Fermi energy.

Under no thermal gradients and external force, the Fermi surface is a circle in 2D k -space.

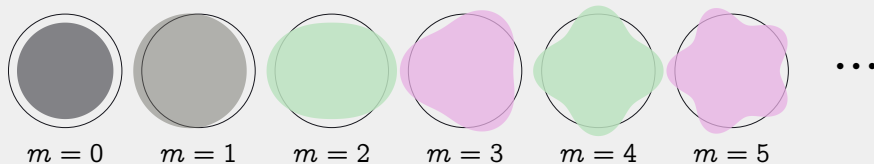


Figure 3: CIRCULAR HARMONICAL PERTURBATIONS OF THE FERMI SURFACE

The Fermi surface may be perturbed by external fields. Since a perturbation has to be smooth, we can expand the perturbation in terms of circular harmonics $e^{im\theta}$ in 2D.

1.3

Boltzmann equation

The Boltzmann equation describes the time evolution of the distribution function $f(t, \mathbf{p})$ in phase space:

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{x}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{x}, \mathbf{p})] \quad 2$$

- $\frac{\partial f}{\partial t}$ is the time evolution of the distribution function
- $\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}}$ is the advection term (by, for example, thermal gradients)
- $\mathbf{F} \cdot \frac{\partial f}{\partial \hbar \mathbf{p}}$ is the external force term
- $\hat{\mathcal{J}}[f]$ is the collision integral, which describes the effect of collisions on the distribution function

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Recovering equations

2.1

Linearization of the collision integral

Ideas

- Consider **only perturbation near Fermi's equilibrium**

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

- Transform the integrals into a linear operator

Preparation

Fermi equilibrium:
$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \quad 3$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \quad 4$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \quad 5$$

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

7

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

8

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

7

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

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Perturbation of interest: $\delta \mu = T \psi$ (temperature dependent change in chemical potential)

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T \psi \quad 9$$

$$= \left(-\frac{1}{T} f_0 (1 - f_0) \right) \times T \psi \quad 10$$

$$= f_0 (1 - f_0) \psi \quad 11$$

Collision integral

$$\hat{\mathcal{J}}[f] = - \int \dots \int \frac{1}{(2\pi)^6} d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B}, \quad 12$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'}) (1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1) (1 - f_2), \quad 13$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \quad 14$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0 :

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \quad 15$$

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The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1} \right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2} \right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'} \right)_{f_0} \delta f_1' + \left(\frac{\partial \mathcal{B}}{\partial f_2'} \right)_{f_0} \delta f_2' \quad 17$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}), \quad 18$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1}f_{0,2}(1 - f_{0,1}), \quad 19$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,2'}) - f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}), \quad 20$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,1'}) - f_{0,1'}(1 - f_{0,1})(1 - f_{0,2}). \quad 21$$

Substitute $\delta f_i = f_{0,i}(1 - f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \left[f_{0,2}(1 - f_{0,1'}) (1 - f_{0,2'}) + f_{0,1'} f_{0,2'} (1 - f_{0,2}) \right] \times f_{0,1}(1 - f_{0,1})\psi_1 \quad 22$$

$$= \left[f_{0,1} f_{0,2} (1 - f_{0,1'}) (1 - f_{0,2'}) (1 - f_{0,1}) + f_{0,1} f_{0,1'} f_{0,2'} (1 - f_{0,1}) (1 - f_{0,2}) \right] \psi_1 \quad 23$$

$$= \left(f_{0,1} f_{0,2} (1 - f_{0,1'}) (1 - f_{0,2'}) \right) \left[(1 - f_{0,1}) + f_{0,1} \right] \psi_1 \quad 24$$

$$= \left(f_{0,1} f_{0,2} (1 - f_{0,1'}) (1 - f_{0,2'}) \right) \psi_1 \quad 25$$

Define $F_{121'2'} = f_{0,1} f_{0,2} (1 - f_{0,1'}) (1 - f_{0,2'}) = f_{0,1'} f_{0,2'} (1 - f_{0,1}) (1 - f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'} \psi_1 \quad 26$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'} \psi_i \quad 27$$

Hence,

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_1' + \left(\frac{\partial \mathcal{B}}{\partial f_2'}\right)_{f_0} \delta f_2' \quad 28$$

$$= F_{121'2'} [\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}] \quad 29$$

$$= F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \quad 30$$

$$\hat{\mathcal{T}}[f] = -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \quad 31$$

$$= -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) (\mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}}) \quad 32$$

- $\hat{\mathcal{T}}[f_0] = 0$: Collision integral vanishes at equilibrium
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\begin{aligned} \hat{\mathcal{T}}[f] = & -\frac{1}{(2\pi^6)} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 \\ & \times W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 33 \end{aligned}$$

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Odd-Even effect